

On a Polytope
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On a Polytope Associated to Products of Algebraic Integer Conjugates

Greg Knapp

University of Calgary

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Pacific Institute *for the*
Mathematical Sciences

Joint work with Seda Albayrak, Samprit Ghosh, and Khoa Nguyen.

Land Acknowledgment

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The University of Calgary, located in the heart of Southern Alberta, both acknowledges and pays tribute to the traditional territories of the peoples of Treaty 7, which include the Blackfoot Confederacy (comprised of the Siksika, the Piikani, and the Kainai First Nations), the Tsuut'ina First Nation, and the Stoney Nakoda (including Chiniki, Bearspaw, and Goodstoney First Nations). The City of Calgary is also home to the Métis Nation of Alberta Districts 5 and 6.

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Theorem (Bugead and Nguyen, 2023)

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Theorem (Bugead and Nguyen, 2023)

Let ξ be an irrational, algebraic number of degree $d \geq 3$. Let $\varepsilon > 0$. Let $(u_n)_{n \geq 1}$ be a non-degenerate linear recurrence sequence of rational integers which is not a polynomial sequence. Then there are only finitely many u_n for which there exists a $v_n \in \mathbb{Z}$ so that

$$\left| \xi - \frac{v_n}{u_n} \right| < \frac{1}{|u_n|^{1 + \frac{1}{d-1} + \varepsilon}}.$$

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Interpretation

Certain sequences cannot serve as denominators for good rational approximations of ξ .

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Let ξ be an irrational, algebraic number of degree $d \geq 3$. Let $\varepsilon > 0$. Let $(u_n)_{n \geq 1}$ be a non-degenerate linear recurrence sequence of rational integers which is not a polynomial sequence. Then there are only finitely many u_n for which there exists a $v_n \in \mathbb{Z}$ so that

$$\left| \xi - \frac{v_n}{u_n} \right| < \frac{1}{|u_n|^{1 + \frac{1}{d-1} + \varepsilon}}.$$

Question

Can the exponent of $\frac{1}{d-1}$ be improved (decreased) at all in order to achieve the same result?

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Notation

For the rest of this talk, whenever we write $f(x)$, it will have the following properties and we will use the following notation:

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Notation

For the rest of this talk, whenever we write $f(x)$, it will have the following properties and we will use the following notation:

■ $f(x) \in \mathbb{Z}[x]$.

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For the rest of this talk, whenever we write $f(x)$, it will have the following properties and we will use the following notation:

- $f(x) \in \mathbb{Z}[x]$.
- $f(x)$ is monic and irreducible.

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- $f(x) \in \mathbb{Z}[x]$.
- $f(x)$ is monic and irreducible.
- d will denote the degree of $f(x)$.

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Notation

For the rest of this talk, whenever we write $f(x)$, it will have the following properties and we will use the following notation:

- $f(x) \in \mathbb{Z}[x]$.
- $f(x)$ is monic and irreducible.
- d will denote the degree of $f(x)$.
- $f(x)$ has complex roots $\alpha_0, \dots, \alpha_{d-1}$ which satisfy

$$|\alpha_0| \geq |\alpha_1| \geq \dots \geq |\alpha_{d-1}|.$$

Source of the Exponent

The exponent in Bugueaud and Nguyen's theorem comes from the following fact:

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The exponent in Bugueaud and Nguyen's theorem comes from the following fact:

Fact

If $f(x) \in \mathbb{Z}[x]$ is monic and irreducible of degree d , then

$$|\alpha_0||\alpha_1|^{d-1} \geq 1.$$

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Proof

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Proof

$$|\alpha_0||\alpha_1|^{d-1} = |\alpha_0||\alpha_1||\alpha_1|^{d-2}$$

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$$|\alpha_0||\alpha_1|^{d-1} \geq 1.$$

Proof

$$|\alpha_0||\alpha_1|^{d-1} = |\alpha_0||\alpha_1||\alpha_1|^{d-2} \geq |\alpha_0||\alpha_1||\alpha_2|^{d-2}$$

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Proof

$$\begin{aligned} |\alpha_0||\alpha_1|^{d-1} &= |\alpha_0||\alpha_1||\alpha_1|^{d-2} \geq |\alpha_0||\alpha_1||\alpha_2|^{d-2} \\ &\geq |\alpha_0||\alpha_1| \cdots |\alpha_{d-1}| \end{aligned}$$

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Proof

$$\begin{aligned} |\alpha_0||\alpha_1|^{d-1} &= |\alpha_0||\alpha_1||\alpha_1|^{d-2} \geq |\alpha_0||\alpha_1||\alpha_2|^{d-2} \\ &\geq |\alpha_0||\alpha_1| \cdots |\alpha_{d-1}| = |f(0)| \geq 1. \end{aligned}$$

Question

Can we replace $d - 1$ by anything else and still have the fact be true?

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Question

Let $d \geq 2$ be an integer. For which values of $c \geq 0$ is it true that for every monic, irreducible $f(x) \in \mathbb{Z}[x]$ of degree d ,

$$|\alpha_0||\alpha_1|^c \geq 1?$$

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$$|\alpha_0||\alpha_1|^c \geq 1?$$

Partial Answer

If $0 \leq c \leq d - 1$, then the above property holds:

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$$|\alpha_0| |\alpha_1|^c \geq 1?$$

Partial Answer

If $0 \leq c \leq d - 1$, then the above property holds:

- Pick an appropriate $f(x)$.

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$$|\alpha_0||\alpha_1|^c \geq 1?$$

Partial Answer

If $0 \leq c \leq d - 1$, then the above property holds:

- Pick an appropriate $f(x)$.
- If $|\alpha_1| \geq 1$, then $|\alpha_0||\alpha_1|^c \geq 1$.

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Partial Answer

If $0 \leq c \leq d - 1$, then the above property holds:

- Pick an appropriate $f(x)$.
- If $|\alpha_1| \geq 1$, then $|\alpha_0||\alpha_1|^c \geq 1$.
- Otherwise, $|\alpha_1| < 1$, so

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- Otherwise, $|\alpha_1| < 1$, so

$$|\alpha_0||\alpha_1|^c \geq |\alpha_0||\alpha_1|^{d-1}$$

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$$|\alpha_0| |\alpha_1|^c \geq 1?$$

Deeper fact

If the above property holds, then $c \leq d - 1$.

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Deeper fact

If the above property holds, then $c \leq d - 1$.

■ Why?

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$$|\alpha_0| |\alpha_1|^c \geq 1?$$

Deeper fact

If the above property holds, then $c \leq d - 1$.

- Why?
- Let's look at the family of polynomials
$$f_{d,1,h}(x) = x^d - hx^{d-1} - 1.$$

A Useful Example

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Definition

For any integers h and d with $d \geq 2$, let

$$f_{d,1,h}(x) = x^d - hx^{d-1} - 1.$$

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For any integers h and d with $d \geq 2$, let

$$f_{d,1,h}(x) = x^d - hx^{d-1} - 1.$$

Facts

- For infinitely many integers h , the polynomial $f_{d,1,h}(x)$ is irreducible over $\mathbb{Z}[x]$.

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$$f_{d,1,h}(x) = x^d - hx^{d-1} - 1.$$

Facts

- For infinitely many integers h , the polynomial $f_{d,1,h}(x)$ is irreducible over $\mathbb{Z}[x]$.
- $f_{d,1,h}$ has one “large” root: $|\alpha_0| \asymp |h|$.

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For any integers h and d with $d \geq 2$, let

$$f_{d,1,h}(x) = x^d - hx^{d-1} - 1.$$

Facts

- For infinitely many integers h , the polynomial $f_{d,1,h}(x)$ is irreducible over $\mathbb{Z}[x]$.
- $f_{d,1,h}$ has one “large” root: $|\alpha_0| \asymp |h|$.
- $f_{d,1,h}$ has $d - 1$ “small” roots:

$$|\alpha_1|, \dots, |\alpha_{d-1}| \asymp |h|^{-1/(d-1)}.$$

$d - 1$ Is The Best Possible Exponent

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Claim

Suppose that $c \geq 0$ has the property that for every monic, irreducible $f(x) \in \mathbb{Z}[x]$ of degree d ,

$$|\alpha_0| |\alpha_1|^c \geq 1.$$

Then $c \leq d - 1$.

$d - 1$ Is The Best Possible Exponent

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Main Theorem

Claim

Suppose that $c \geq 0$ has the property that for every monic, irreducible $f(x) \in \mathbb{Z}[x]$ of degree d ,

$$|\alpha_0| |\alpha_1|^c \geq 1.$$

Then $c \leq d - 1$.

Proof

Apply this property to each of the (infinitely many) irreducible polynomials of the form $f_{d,1,h}(x) = x^d - hx^{d-1} - 1$:

$d - 1$ Is The Best Possible Exponent

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Recap

$c \in [0, d - 1]$ if and only if $c \geq 0$ and for every irreducible, monic $f(x) \in \mathbb{Z}[x]$,

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Corollary

The exponent in our motivating theorem is optimal.

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Follow-Up Questions

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Follow-Up Questions

- If this is the “one-dimensional problem,” what do the higher-dimensional problems look like?

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Corollary

The exponent in our motivating theorem is optimal.

Follow-Up Questions

- If this is the “one-dimensional problem,” what do the higher-dimensional problems look like?
- For $c \in [0, d - 1]$, can we guarantee that $|\alpha_0||\alpha_1|^c > 1$? If so, by how much?

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Let $d \geq 2$ be an integer and let $1 \leq k < d$ be another integer.

Problem Statement

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Let $d \geq 2$ be an integer and let $1 \leq k < d$ be another integer. Let $E_{k,d} \subseteq (\mathbb{R}_{\geq 0})^k$ be the set of all tuples $(c_1, \dots, c_k)$ such that

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$$|\alpha_0| |\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} \geq 1.$$

Example

$E_{1,d}$ is the set of all $c \in \mathbb{R}_{\geq 0}$ for which $|\alpha_0| |\alpha_1|^c \geq 1$ for all $f(x)$.

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Example

$E_{1,d}$ is the set of all $c \in \mathbb{R}_{\geq 0}$ for which $|\alpha_0| |\alpha_1|^c \geq 1$ for all $f(x)$.

Example

$E_{2,d}$ is the set of all $(c_1, c_2) \in (\mathbb{R}_{\geq 0})^2$ for which $|\alpha_0| |\alpha_1|^{c_1} |\alpha_2|^{c_2} \geq 1$ for all $f(x)$.

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Question

What is the shape of $E_{k,d}$?

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Question

What is the shape of $E_{k,d}$?

Initial Facts

$E_{k,d}$ is convex and bounded.

The Easier Part Becomes More Difficult

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Main Theorem

$k = 1$

It was easy to show that

$$|\alpha_0| |\alpha_1|^c \geq 1$$

holds for $c \leq d - 1$.

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Main Theorem

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It was easy to show that

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$k > 1$

When looking at the inequality

$$|\alpha_0||\alpha_1|^{c_1}|\alpha_2|^{c_2} \cdots |\alpha_k|^{c_k} \geq 1,$$

it is no longer clear what values of c_i we should be checking to make our job “easy.”

The Difficult Part Doesn't Become More Difficult

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Main Theorem

$$k = 1$$

We used the family of polynomials $f_{d,1,h}(x) = x^d - hx^{d-1} - 1$ to show that if

$$|\alpha_0||\alpha_1|^c \geq 1$$

holds, then $c \leq d - 1$.

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$k > 1$

Let's use similar polynomials to find similar restrictions on $E_{k,d}$.

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A Restriction on $E_{k,d}$

- Suppose $(c_1, \dots, c_k) \in E_{k,d}$, so that for any monic, irreducible $f(x) \in \mathbb{Z}[x]$, we have $|\alpha_0| |\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} \geq 1$.

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- Hence, $1 - \frac{1}{d-1} \sum_{j=1}^k c_j \geq 0$, i.e. $\sum_{j=1}^k c_j \leq d - 1$.

Takeaway

$$E_{k,d} \subseteq \{(c_1, \dots, c_k) \in (\mathbb{R}_{\geq 0})^k : c_1 + \cdots + c_k \leq d - 1\}$$

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Question

Is

$$E_{k,d} = \{(c_1, \dots, c_k) \in (\mathbb{R}_{\geq 0})^k : c_1 + \dots + c_k \leq d - 1\}?$$

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$k = 1$

Yes! $E_{1,d} = [0, d - 1]$.

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$k = 1$

Yes! $E_{1,d} = [0, d - 1]$.

$k > 1$

No. The point $(0, \dots, 0, d - 1)$ is not in $E_{k,d}$ despite satisfying $c_1 + \dots + c_k \leq d - 1$.

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More Restrictions on $E_{k,d}$

- Suppose $(c_1, \dots, c_k) \in E_{k,d}$, so that for any monic, irreducible $f(x) \in \mathbb{Z}[x]$, we have $|\alpha_0||\alpha_1|^{c_1} \dots |\alpha_k|^{c_k} \geq 1$.

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More Restrictions on $E_{k,d}$

- Suppose $(c_1, \dots, c_k) \in E_{k,d}$, so that for any monic, irreducible $f(x) \in \mathbb{Z}[x]$, we have $|\alpha_0||\alpha_1|^{c_1} \dots |\alpha_k|^{c_k} \geq 1$.
- For $1 \leq i \leq k$, let $f_{d,i,h}(x) = x^d - hx^{d-i} - 1$, which has

$$|\alpha_0|, \dots, |\alpha_{i-1}| \asymp |h|^{1/i} \text{ and } |\alpha_i|, \dots, |\alpha_{d-1}| \asymp |h|^{-\frac{1}{d-i}}.$$

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- Apply the fact that $(c_1, \dots, c_k) \in E_{k,d}$:

$$1 \leq |\alpha_0||\alpha_1|^{c_1} \dots |\alpha_k|^{c_k}.$$

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- Apply the fact that $(c_1, \dots, c_k) \in E_{k,d}$:

$$1 \leq |\alpha_0||\alpha_1|^{c_1} \dots |\alpha_k|^{c_k} \asymp |h|^{i^{-1}(1 + \sum_{j=1}^{i-1} c_j) - \frac{1}{d-i} \sum_{j=i}^k c_j}.$$

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- Apply the fact that $(c_1, \dots, c_k) \in E_{k,d}$:

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- Hence,

$$-\frac{d-i}{i} \sum_{j=1}^{i-1} c_j + \sum_{j=i}^k c_j \leq \frac{d-i}{i}.$$

Summary of Restrictions

Summary

We have found that every point $(c_1, \dots, c_k) \in E_{k,d}$ satisfies

$$c_i \geq 0 \quad \text{for } 1 \leq i \leq k$$

$$-\frac{d-i}{i} \sum_{j=1}^{i-1} c_j + \sum_{j=i}^k c_j \leq \frac{d-i}{i} \quad \text{for } 1 \leq i \leq k$$

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Summary

We have found that every point $(c_1, \dots, c_k) \in E_{k,d}$ satisfies

$$\begin{aligned} c_i &\geq 0 && \text{for } 1 \leq i \leq k \\ -\frac{d-i}{i} \sum_{j=1}^{i-1} c_j + \sum_{j=i}^k c_j &\leq \frac{d-i}{i} && \text{for } 1 \leq i \leq k \end{aligned}$$

Question

Is this enough restrictions?

Reformulating the Question

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Main Theorem

Definition

Let $P_{k,d}$ be the set of all $(c_1, \dots, c_k) \in \mathbb{R}^k$ which satisfy

$$c_i \geq 0 \quad \text{for } 1 \leq i \leq k$$

$$-\frac{d-i}{i} \sum_{j=1}^{i-1} c_j + \sum_{j=i}^k c_j \leq \frac{d-i}{i} \quad \text{for } 1 \leq i \leq k$$

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Equivalent Question

We know $E_{k,d} \subseteq P_{k,d}$. Is $E_{k,d} = P_{k,d}$?

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Equivalent Question

We know $E_{k,d} \subseteq P_{k,d}$. Is $E_{k,d} = P_{k,d}$?

Note

$P_{k,d}$ is a polytope.

Important Reduction

Question

We know $E_{k,d} \subseteq P_{k,d}$. Is $E_{k,d} = P_{k,d}$?

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Question

We know $E_{k,d} \subseteq P_{k,d}$. Is $E_{k,d} = P_{k,d}$?

A Reduction

Since $E_{k,d}$ is convex and $P_{k,d}$ is a polytope, it suffices to check if the vertices of $P_{k,d}$ lie in $E_{k,d}$.

Important Reduction

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A Reduction

Since $E_{k,d}$ is convex and $P_{k,d}$ is a polytope, it suffices to check if the vertices of $P_{k,d}$ lie in $E_{k,d}$.

Question

What are the vertices of $P_{k,d}$?

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Theorem (Albayrak, Ghosh, K., Nguyen)

$P_{k,d}$ is the polytope whose vertices are in bijection with the subsets of $\{1, \dots, k\}$ as follows:

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Theorem (Albayrak, Ghosh, K., Nguyen)

$P_{k,d}$ is the polytope whose vertices are in bijection with the subsets of $\{1, \dots, k\}$ as follows:

- *For each subset $I \subseteq \{1, \dots, k\}$, write $I = \{i_1, \dots, i_n\}$ where $i_1 < i_2 < \dots < i_n$.*

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$P_{k,d}$ is the polytope whose vertices are in bijection with the subsets of $\{1, \dots, k\}$ as follows:

- For each subset $I \subseteq \{1, \dots, k\}$, write $I = \{i_1, \dots, i_n\}$ where $i_1 < i_2 < \dots < i_n$.
- Then $(c_1, \dots, c_k)$ is a vertex where, for each $1 \leq j \leq k$,

$$c_j = \begin{cases} 0 & \text{if } j \notin I \\ \frac{i_{\ell+1} - i_\ell}{i_1} & \text{if } j = i_\ell \text{ for some } \ell < n \\ \frac{d - i_n}{i_1} & \text{if } j = i_n \end{cases}$$

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Recall

The subset $I = \{i_1, \dots, i_n\} \subseteq \{1, \dots, k\}$ produces the vertex $(c_1, \dots, c_k)$ where

$$c_j = \begin{cases} 0 & \text{if } j \notin I \\ \frac{i_{\ell+1} - i_\ell}{i_1} & \text{if } j = i_\ell \text{ for some } \ell < n \\ \frac{d - i_n}{i_1} & \text{if } j = i_n \end{cases}$$

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Example

The vertices of $P_{2,d}$ are:

I	Vertex

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Example

The vertices of $P_{2,d}$ are:

I	Vertex
$\emptyset$	$(0, 0)$

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Example

The vertices of $P_{2,d}$ are:

I	Vertex
$\emptyset$	$(0, 0)$
$\{1\}$	$(d - 1, 0)$
$\{2\}$	$(0, (d - 2)/2)$

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$\{1, 2\}$	$(1, d - 2)$

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The Final Steps

- We know $E_{k,d}$ is a convex subset of $P_{k,d}$.

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The Final Steps

- We know $E_{k,d}$ is a convex subset of $P_{k,d}$.
- We know that $P_{k,d}$ is a polytope and we can identify its vertices.

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The Final Steps

- We know $E_{k,d}$ is a convex subset of $P_{k,d}$.
- We know that $P_{k,d}$ is a polytope and we can identify its vertices.
- To see if $E_{k,d} = P_{k,d}$, it suffices to check that the vertices of $P_{k,d}$ are in $E_{k,d}$.

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Theorem (Albayrak, Ghosh, K., Nguyen)

$$E_{k,d} = P_{k,d}.$$

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$E_{k,d}$ is the set of all points $(c_1, \dots, c_k) \in \mathbb{R}^k$ which satisfy

$$c_i \geq 0 \quad \text{for } 1 \leq i \leq k$$

$$-\frac{d-i}{i} \sum_{j=1}^{i-1} c_j + \sum_{j=i}^k c_j \leq \frac{d-i}{i} \quad \text{for } 1 \leq i \leq k$$

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Example

$E_{1,d} \subseteq \mathbb{R}$ is defined by the inequalities

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$$c \geq 0$$

$$c \leq d - 1$$

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Main Theorem

Theorem (Albayrak, Ghosh, K., Nguyen)

$E_{k,d}$ is the set of all points $(c_1, \dots, c_k) \in \mathbb{R}^k$ which satisfy

$$c_i \geq 0 \quad \text{for } 1 \leq i \leq k$$

$$-\frac{d-i}{i} \sum_{j=1}^{i-1} c_j + \sum_{j=i}^k c_j \leq \frac{d-i}{i} \quad \text{for } 1 \leq i \leq k$$

Example

$E_{2,d} \subseteq \mathbb{R}$ is defined by the inequalities

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Example

$E_{2,d} \subseteq \mathbb{R}$ is defined by the inequalities $x \geq 0, y \geq 0$, and

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Example

$E_{2,d} \subseteq \mathbb{R}$ is defined by the inequalities $x \geq 0, y \geq 0$, and

$$x + y \leq d - 1 \quad (i = 1)$$

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Example

$E_{2,d} \subseteq \mathbb{R}$ is defined by the inequalities $x \geq 0, y \geq 0$, and

$$x + y \leq d - 1 \quad (i = 1)$$

$$-\frac{d-2}{2}x + y \leq \frac{d-2}{2} \quad (i = 2).$$

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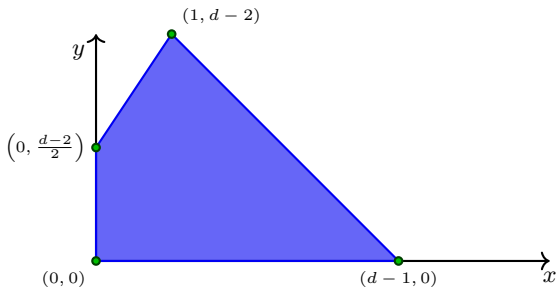
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A picture of $E_{2,d}$ created in SageMath:

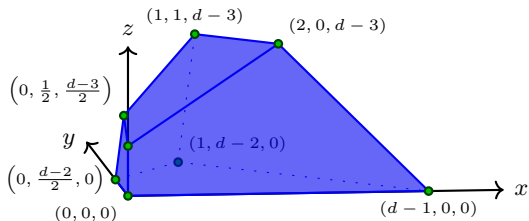


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An image of $E_{3,d}$ created in SageMath:



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For $c \in [0, d - 1]$, is it possible that

$$|\alpha_0||\alpha_1|^c = 1?$$

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Main Theorem

Question

For $c \in [0, d - 1]$, is it possible that

$$|\alpha_0||\alpha_1|^c = 1?$$

“Trivial” Answer

If $f(x)$ is cyclotomic, then

$$|\alpha_0||\alpha_1|^c = 1$$

for any c .

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Reduction

If $f(x)$ is not cyclotomic, then $|\alpha_0||\alpha_1|^c = 1$ only if $c = d - 1$.

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Question

Is it possible that

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Nontrivial Answers

■ $f(x) = x^2 - x - 1$ has

$$|\alpha_0||\alpha_1|^{d-1} = |\alpha_0\alpha_1|$$

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Nontrivial Answers

- $f(x) = x^2 - x - 1$ has

$$|\alpha_0||\alpha_1|^{d-1} = |\alpha_0\alpha_1| = |f(0)|$$

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Nontrivial Answers

- $f(x) = x^2 - x - 1$ has

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- If $f(x) \in \mathbb{Z}[x]$ is a monic, irreducible cubic with $|f(0)| = 1$ and its two smaller roots are complex conjugates, then

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- $f(x) = x^3 + x^2 - x + 1$ is such a polynomial.

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- $f(x) = x^3 + x^2 - x + 1$ is such a polynomial.
- If $\deg(f) > 3$, then

$$|\alpha_0||\alpha_1|^{d-1} > 1.$$

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Theorem (Albayrak, Ghosh, K., Nguyen)

If $d > 3k$ and $(c_1, \dots, c_k) \in E_{k,d}$, then any monic, irreducible, noncyclotomic $f(x) \in \mathbb{Z}[x]$ has

$$|\alpha_0| |\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} > 1.$$

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Note

The condition $d > 3k$ is necessary.

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$$|\alpha_0| |\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} > 1.$$

Question

If $d > 3k$, can we get a lower bound on

$$|\alpha_0| |\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} - 1?$$

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Question

What should a lower bound on $|\alpha_0||\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} - 1$ look like?

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Question

What should a lower bound on $|\alpha_0||\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} - 1$ look like?

The “zero-dimensional” version of this problem is the Schinzel-Zassenhaus conjecture:

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Question

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Theorem (Dimitrov)

$$|\alpha_0| - 1 \gg \frac{1}{d}.$$

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Question

Is it possible to get a lower bound on $|\alpha_0||\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} - 1$ that only depends on the degree d ?

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Question

Is it possible to get a lower bound on $|\alpha_0||\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} - 1$ that only depends on the degree d ?

Answer

■ No.

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Question

Is it possible to get a lower bound on $|\alpha_0||\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} - 1$ that only depends on the degree d ?

Answer

- No.
- Use more precise estimates for the roots of $f_{d,1,h}(x) = x^d - hx^{d-1} - 1$ to find that

$$|\alpha_0||\alpha_1|^{d-1} - 1 \ll \frac{1}{|h|}.$$

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$$|\alpha_0||\alpha_1|^{d-1} - 1 \ll \frac{1}{|h|}.$$

- Instead, our lower bound will need to depend on the *height* of our polynomial.

Lower Bounds on Power Products

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Notation

Let $H(f)$ denote the maximum of the absolute values of the coefficients of $f(x)$.

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Main Theorem

Notation

Let $H(f)$ denote the maximum of the absolute values of the coefficients of $f(x)$.

Theorem (Albayrak, Ghosh, K., Nguyen)

Let $d > 3k$. Then there exist effectively computable constants $A_1, A_2 > 0$ so that for any $(c_1, \dots, c_k) \in E_{k,d}$ and any monic, irreducible, noncyclotomic $f(x) \in \mathbb{Z}[x]$ of degree d ,

$$|\alpha_0| |\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k} - 1 \geq A_1 (1 + H(f))^{-A_2 k d^3}.$$

If $d > 4k$, then d^3 can be replaced with d^2 .

Note

A_1 depends on d , A_2 is absolute.

Thank you!

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Questions?